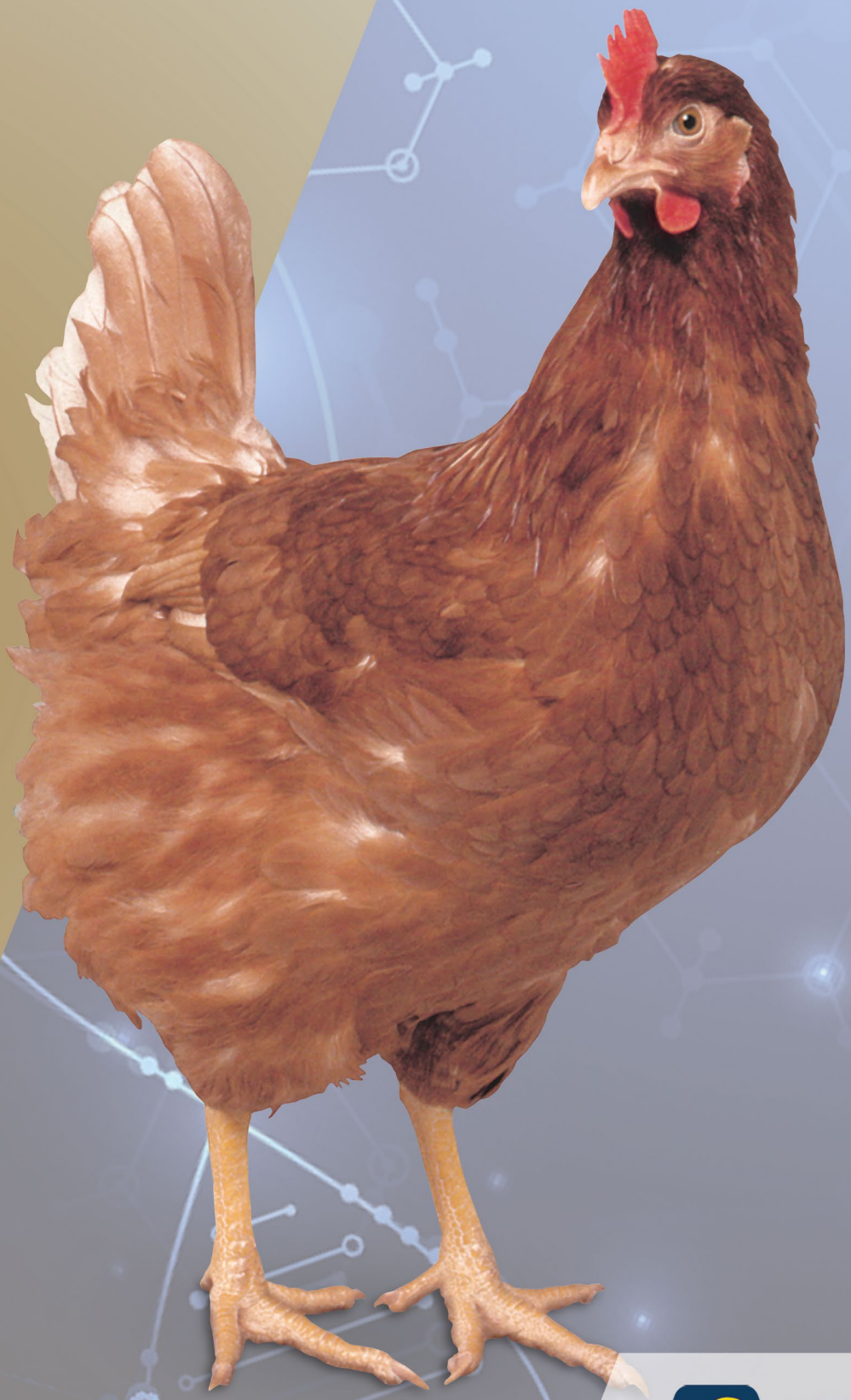
Hy-Line® BROWN

PERFORMANCE STANDARDS GUIDE

COMMERCIAL LAYERS

CONVENTIONAL SYSTEMS



Hy-Line®

Use of the Performance Guide

The genetic potential of Hy-Line Brown Commercial can only be realized if good poultry husbandry practices and management are used. This management guide outlines successful flock management programmes for Hy-Line Variety Brown Commercial based on field experience compiled by Hy-Line International and using an extensive commercial layer flock database of Hy-Line flocks from all parts of the world. Hy-Line International Management Guides are periodically updated as new performance data and/or nutrition information become available.

The information and suggestions contained in this management guide should be used for guidance and educational purposes only, recognising that local environmental and disease conditions may vary and a guide cannot cover all possible circumstances. While every attempt has been made to ensure that the information presented is accurate and reliable at the time of publication, Hy-Line International cannot accept responsibility for any errors, omissions or inaccuracies in such information or management suggestions. Further, Hy-Line International does not warrant or make any representations or guarantees regarding the use, validity, accuracy, or reliability of, or flock performance or productivity resulting from the use of, or otherwise respecting, such information or management suggestions. In no event shall Hy-Line International be liable for any special, indirect or consequential damages or special damages whatsoever arising out of or in connection with the use of the information or management suggestions contained in this management guide.

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Hy-Line Brown
Online Management Guide

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Summary of Performance Standards

REARING PERIOD (TO 17 WEEKS):	
Livability	98%
Feed Consumed	5.99 kg
Body Weight at 17 Weeks	1357–1434 g
LAYING PERIOD (TO 100 WEEKS):	
Percent Peak	93.6–98.5%
Hen-Day Eggs to 60 Weeks Hen-Day Eggs to 100 Weeks	259.8–274.7 481.8–514.5
Hen-Housed Eggs to 60 Weeks Hen-Housed Eggs to 100 Weeks	256.2–270.8 466.4–497.9
Livability to 60 Weeks Livability to 100 Weeks	97.17% 91.40%
Days to 50% Production (from hatch)	139
Egg Weight at 26 Weeks Egg Weight at 32 Weeks Egg Weight at 70 Weeks Egg Weight at 100 Weeks	55.2–58.6 g 58.9–62.6 g 61.2–65.0 g 62.2–66.1 g
Total Egg Mass per Hen-Housed (18–80 weeks) Total Egg Mass per Hen-Housed (18–100 weeks)	23.3 kg 29.8 kg
Body Weight at 26 Weeks Body Weight at 32 Weeks Body Weight at 70 Weeks Body Weight at 100 Weeks	1810–1913 g 1876–1984 g 1954–2066 g 1965–2078 g
Freedom From Egg Inclusions	Excellent
Shell Strength	Excellent
Shell Color Score at 38 Weeks Shell Color Score at 56 Weeks Shell Color Score at 72 Weeks Shell Color Score at 90 Weeks	87 85 81 79
Haugh Units at 38 Weeks Haugh Units at 56 Weeks Haugh Units at 72 Weeks Haugh Units at 90 Weeks	90.0 84.0 81.0 79.7
Average Daily Feed Consumption (19–100 weeks)	109.1–116.9 g/day per bird
Feed Conversion Rate, kg Feed/kg Eggs (20–60 weeks) Feed Conversion Rate, kg Feed/kg Eggs (20–100 weeks)	2.06–2.15 2.15–2.24
Feed Utilization, kg Egg/kg Feed (20–60 weeks) Feed Utilization, kg Egg/kg Feed (20–100 weeks)	0.47–0.48 0.45–0.47
Feed Consumption per 10 Eggs (20–60 weeks) Feed Consumption per 10 Eggs (20–100 weeks)	1.23–1.28 kg 1.29–1.34 kg
Feed Consumption per Dozen Eggs (20–60 weeks) Feed Consumption per Dozen Eggs (20–100 weeks)	1.48–1.54 kg 1.55–1.61 kg
Skin Color	Yellow
Condition of Droppings	Dry

Rearing Period Performance Table

AGE (weeks)	MORTALITY Cumulative (%)	BODY WEIGHT (g)	WATER INTAKE (ml/bird/day)	FEED INTAKE (g/bird/day)	CUMULATIVE FEED INTAKE (g/bird to date)	UNIFORMITY %
1	0.40	69 – 73	24 – 33	16 – 17	110 – 117	
2	0.55	119 – 126	27 – 37	18 – 19	234 – 248	>85%
3	0.65	186 – 197	31 – 45	21 – 23	378 – 406	
4	0.75	266 – 281	38 – 55	26 – 28	557 – 599	
5	**0.85**	**357 – 377**	**47 – 67**	**31 – 33**	**777 – 832**	>80%
6	0.95	456 – 482	56 – 81	37 – 40	1039 – 1114	
7	1.05	561 – 593	65 – 92	43 – 46	1341 – 1438	
8	1.15	668 – 706	72 – 104	48 – 52	1679 – 1802	
9	1.25	772 – 816	80 – 114	53 – 57	2050 – 2201	
10	**1.35**	**871 – 921**	**86 – 122**	**57 – 61**	**2449 – 2628**	
11	1.45	963 – 1018	91 – 130	61 – 65	2876 – 3082	>85%
12	1.55	1046 – 1105	94 – 136	63 – 68	3316 – 3557	
13	1.63	1120 – 1184	99 – 140	66 – 70	3777 – 4045	
14	1.70	1186 – 1254	102 – 145	68 – 73	4251 – 4554	
15	**1.78**	**1246 – 1317**	**105 – 149**	**70 – 75**	**4740 – 5077**	
16	1.85	1302 – 1377	109 – 155	73 – 78	5249 – 5620	
17	2.00	1357 – 1434	115 – 163	77 – 82	5786 – 6191	>90%

Production Period Performance Table

AGE (weeks)	% HEN-DAY Current	HEN-DAY EGGS Cumulative	HEN-HOUSED EGGS Cumulative	MORT-ALITY Cumulative (%)	BODY WEIGHT (g)	WATER INTAKE (ml/bird/day)	FEED INTAKE (g/bird/day)	HH EGG MASS Cumulative (kg)	AVG. EGG WT. (g/egg)
18	6.7–7.0	0.5–0.5	0.5–0.5	0.24	1411 – 1492	121 – 173	81–86	–	42.8–45.5
19	24.0–25.2	2.2–2.3	2.1–2.2	0.28	1467 – 1551	130 – 185	86–92	0.1 – 0.1	45.2–48.0
20	**52.5–55.0**	**5.8 – 6.1**	**5.8 – 6.1**	**0.32**	**1524 – 1611**	**137 – 197**	**91–98**	**0.3 – 0.3**	**47.6 – 50.5**
21	75.9–79.6	11.1 – 11.7	11.1 – 11.6	0.36	1582 – 1672	144 – 206	96–103	0.5 – 0.6	49.1–52.2
22	87.0–91.2	17.2 – 18.1	17.2 – 18.0	0.40	1638 – 1732	152 – 216	101–108	0.9 – 0.9	50.6–53.7
23	90.9–95.1	23.6 – 24.7	23.5 – 24.6	0.44	1691 – 1788	156 – 224	104–112	1.2 – 1.3	51.9–55.1
24	92.4–97.0	30.1 – 31.5	29.9 – 31.4	0.49	1739 – 1838	161 – 228	107–114	1.6 – 1.6	53.1–56.4
25	**93.4–98.1**	**36.6 – 38.4**	**36.4 – 38.2**	**0.53**	**1779 – 1880**	**162 – 232**	**108–116**	**1.9 – 2.0**	**54.2 – 57.6**
26	93.6–98.5	43.2 – 45.3	43.0 – 45.1	0.58	1810 – 1913	164 – 234	109–117	2.3 – 2.4	55.2–58.6
27	93.6–98.4	49.7 – 52.2	49.5 – 51.9	0.62	1832 – 1937	164 – 235	110–117	2.7 – 2.8	56.1–59.6
28	93.5–98.4	56.2 – 59.0	56.0 – 58.8	0.67	1847 – 1953	165 – 236	110–118	3.1 – 3.2	56.8–60.4
29	93.4–98.3	62.8 – 65.9	62.5 – 65.6	0.72	1857 – 1963	165 – 236	110–118	3.4 – 3.6	57.5–61.0
30	**93.3–98.2**	**69.3 – 72.8**	**68.9 – 72.4**	**0.77**	**1864 – 1971**	**165 – 236**	**110–118**	**3.8 – 4.0**	**58.0 – 61.6**
31	93.2–98.1	75.8 – 79.7	75.4 – 79.2	0.82	1871 – 1977	165 – 236	110–118	4.2 – 4.4	58.5–62.1
32	93.1–98.1	82.3 – 86.5	81.9 – 86.0	0.87	1877 – 1984	165 – 236	110–118	4.6 – 4.8	58.9–62.6
33	92.9–98.0	88.9 – 93.4	88.3 – 92.8	0.92	1883 – 1991	165 – 236	110–118	5.0 – 5.3	59.3–62.9
34	92.8–97.9	95.3 – 100.2	94.7 – 99.6	0.97	1890 – 1998	165 – 236	110–118	5.4 – 5.7	59.5–63.2
35	**92.7–97.8**	**101.8 – 107.1**	**101.2 – 106.4**	**1.03**	**1896 – 2004**	**165 – 236**	**110–118**	**5.8 – 6.1**	**59.7 – 63.4**
36	92.5–97.7	108.3 – 113.9	107.6 – 113.1	1.09	1903 – 2012	165 – 236	110–118	6.2 – 6.5	59.9–63.6
37	92.4–97.6	114.8 – 120.8	114.0 – 119.9	1.14	1909 – 2018	165 – 236	110–118	6.6 – 6.9	60.0–63.8
38	92.3–97.5	121.2 – 127.6	120.3 – 126.6	1.20	1913 – 2023	165 – 236	110–118	7.0 – 7.4	60.1–63.9
39	92.1–97.3	127.7 – 134.4	126.7 – 133.4	1.26	1917 – 2027	165 – 236	110–118	7.4 – 7.8	60.2–63.9
40	**91.9–97.2**	**134.1 – 141.2**	**133.1 – 140.1**	**1.32**	**1921 – 2030**	**165 – 236**	**110–118**	**7.8 – 8.2**	**60.2 – 63.9**
41	91.8–97.1	140.5 – 148.0	139.4 – 146.8	1.38	1924 – 2033	165 – 236	110–118	8.2 – 8.6	60.2–64.0
42	91.6–96.9	147.0 – 154.8	145.7 – 153.5	1.44	1926 – 2036	165 – 236	110–118	8.6 – 9.0	60.3–64.0
43	91.4–96.8	153.4 – 161.6	152.0 – 160.1	1.51	1928 – 2039	165 – 236	110–118	9.0 – 9.4	60.3–64.0
44	91.2–96.6	159.7 – 168.3	158.3 – 166.8	1.57	1930 – 2041	165 – 236	110–118	9.4 – 9.9	60.3–64.1
45	**91.1–96.5**	**166.1 – 175.1**	**164.6 – 173.4**	**1.64**	**1932 – 2043**	**165 – 236**	**110–118**	**9.7 – 10.3**	**60.4 – 64.1**
46	90.9–96.3	172.5 – 181.8	170.8 – 180.1	1.71	1934 – 2044	165 – 236	110–118	10.1 – 10.7	60.4–64.1
47	90.7–96.1	178.8 – 188.5	177.1 – 186.7	1.78	1935 – 2045	165 – 236	110–118	10.5 – 11.1	60.4–64.2
48	90.5–96.0	185.2 – 195.3	183.3 – 193.3	1.85	1936 – 2047	165 – 236	110–118	10.9 – 11.5	60.5–64.2
49	90.2–95.8	191.5 – 202.0	189.5 – 199.8	1.92	1937 – 2048	165 – 236	110–118	11.3 – 11.9	60.5–64.3
50	**90.0–95.6**	**197.8 – 208.7**	**195.6 – 206.4**	**1.99**	**1938 – 2049**	**165 – 236**	**110–118**	**11.7 – 12.3**	**60.5 – 64.3**
51	89.8–95.4	204.1 – 215.3	201.8 – 212.9	2.07	1939 – 2050	165 – 236	110–118	12.1 – 12.7	60.6–64.3
52	89.5–95.2	210.3 – 222.0	207.9 – 219.5	2.15	1940 – 2051	165 – 236	110–118	12.5 – 13.2	60.6–64.4
53	89.3–94.9	216.6 – 228.6	214.0 – 226.0	2.23	1941 – 2052	165 – 236	110–118	12.8 – 13.6	60.6–64.4
54	89.1–94.7	222.8 – 235.3	220.1 – 232.4	2.31	1942 – 2053	165 – 236	110–118	13.2 – 14.0	60.7–64.4
55	**88.8–94.5**	**229.0 – 241.9**	**226.2 – 238.9**	**2.39**	**1943 – 2054**	**165 – 236**	**110–118**	**13.6 – 14.4**	**60.7 – 64.5**
56	88.5–94.2	235.2 – 248.5	232.2 – 245.3	2.47	1944 – 2055	165 – 236	110–118	14.0 – 14.8	60.7–64.5
57	88.3–94.0	241.4 – 255.1	238.3 – 251.7	2.56	1944 – 2056	165 – 236	110–118	14.4 – 15.2	60.8–64.5
58	88.0–93.7	247.6 – 261.6	244.3 – 258.1	2.65	1945 – 2056	165 – 236	110–118	14.7 – 15.6	60.8–64.6

Production Period Performance Table (cont.)

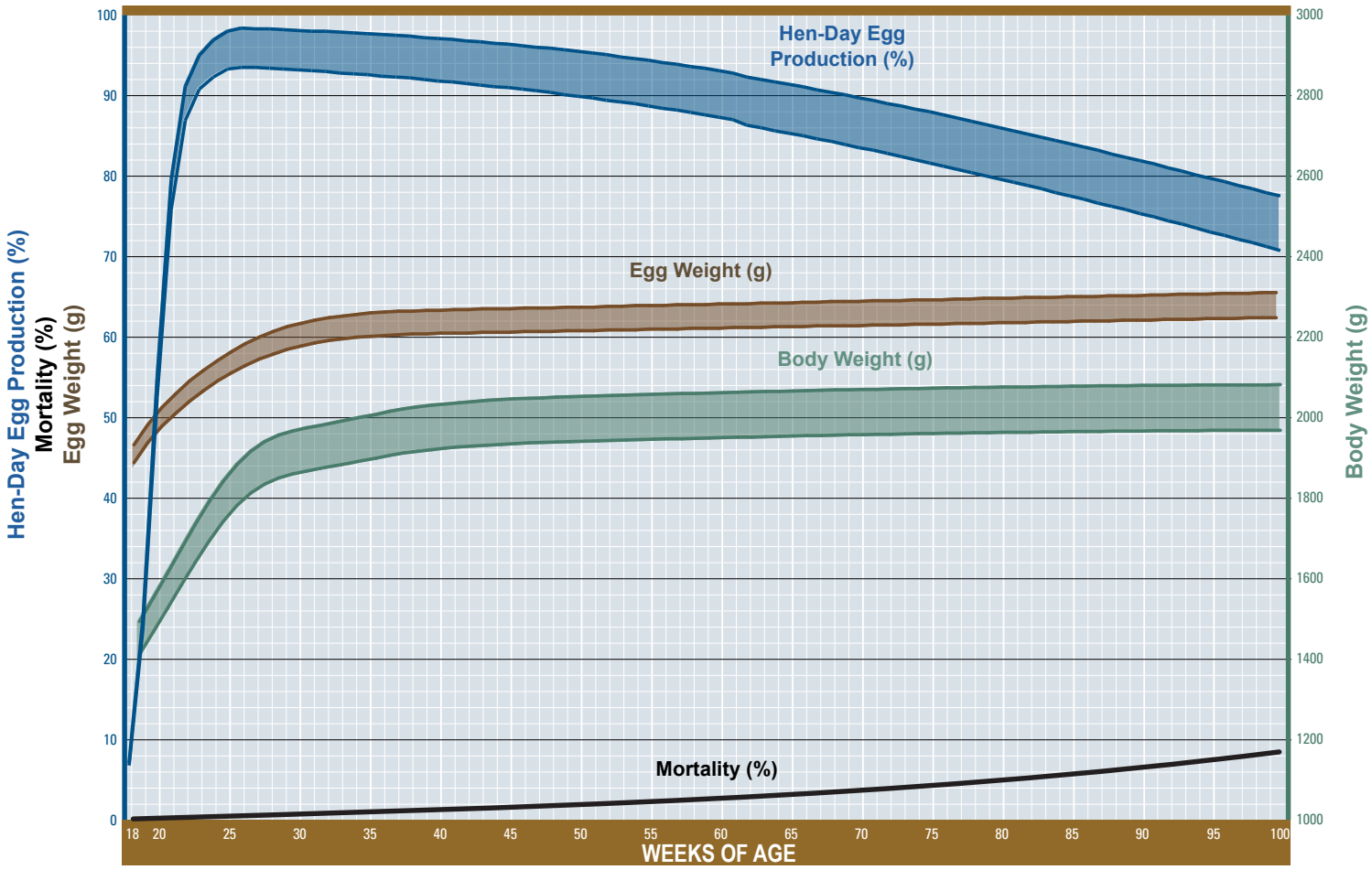
AGE (weeks)	% HEN-DAY Current	HEN-DAY EGGS Cumulative	HEN-HOUSED EGGS Cumulative	MORT-ALITY Cumulative (%)	BODY WEIGHT (g)	WATER INTAKE (ml/bird/day)	FEED INTAKE (g/bird/day)	HH EGG MASS Cumulative (kg)	AVG. EGG WT. (g/egg)
59	87.7–93.5	253.7–268.2	250.2–264.5	2.74	1946–2057	165–236	110–118	15.1–16.0	60.8–64.6
60	**87.4–93.2**	**259.8–274.7**	**256.2–270.8**	**2.83**	**1947–2058**	**165–236**	**110–118**	**15.5–16.4**	**60.9–64.6**
61	87.1–92.9	265.9–281.2	262.1–277.1	2.92	1948–2059	165–236	110–118	15.9–16.8	60.9–64.7
62	86.4–92.4	272.0–287.7	268.0–283.4	3.02	1948–2060	165–236	110–118	16.2–17.2	60.9–64.7
63	86.1–92.1	278.0–294.1	273.8–289.7	3.12	1949–2061	165–236	110–118	16.6–17.6	61.0–64.8
64	85.7–91.8	284.0–300.5	279.6–295.9	3.22	1950–2061	165–236	110–118	17.0–18.0	61.0–64.8
65	**85.4–91.5**	**290.0–306.9**	**285.4–302.1**	**3.32**	**1951–2062**	**165–236**	**110–118**	**17.3–18.4**	**61.0–64.8**
66	85.1–91.2	295.9–313.3	291.2–308.2	3.42	1952–2063	165–236	110–118	17.7–18.7	61.1–64.9
67	84.7–90.8	301.9–319.7	296.9–314.4	3.52	1952–2064	165–236	110–118	18.1–19.1	61.1–64.9
68	84.4–90.5	307.8–326.0	302.6–320.5	3.64	1953–2065	165–236	110–118	18.4–19.5	61.1–64.9
69	84.0–90.2	313.7–332.3	308.2–326.6	3.75	1954–2065	165–236	110–118	18.8–19.9	61.2–65.0
70	**83.6–89.8**	**319.5–338.6**	**313.8–332.6**	**3.86**	**1954–2066**	**165–236**	**110–118**	**19.1–20.3**	**61.2–65.0**
71	83.3–89.5	325.3–344.9	319.4–338.6	3.97	1955–2066	165–236	110–118	19.5–20.7	61.2–65.0
72	82.9–89.1	331.1–351.1	325.0–344.6	4.09	1955–2067	165–236	110–118	19.8–21.0	61.3–65.1
73	82.5–88.8	336.9–357.3	330.5–350.6	4.21	1956–2068	165–236	110–118	20.2–21.4	61.3–65.1
74	82.1–88.4	342.7–363.5	336.0–356.5	4.34	1957–2068	165–236	110–118	20.5–21.8	61.3–65.1
75	**81.7–88.1**	**348.4–369.7**	**341.5–362.4**	**4.46**	**1957–2069**	**165–236**	**110–118**	**20.9–22.2**	**61.4–65.2**
76	81.3–87.7	354.1–375.8	346.9–368.2	4.58	1958–2070	165–236	110–118	21.2–22.5	61.4–65.2
77	80.9–87.3	359.7–381.9	352.3–374.0	4.71	1958–2070	165–236	110–118	21.6–22.9	61.4–65.2
78	80.5–86.9	365.4–388.0	357.7–379.8	4.84	1959–2071	165–236	110–118	21.9–23.3	61.5–65.3
79	80.1–86.5	371.0–394.1	363.0–385.6	4.99	1959–2071	165–236	110–118	22.3–23.6	61.5–65.3
80	**79.7–86.1**	**376.6–400.1**	**368.3–391.3**	**5.13**	**1960–2072**	**165–236**	**110–118**	**22.6–24.0**	**61.5–65.4**
81	79.3–85.7	382.1–406.1	373.6–397.0	5.27	1960–2072	165–236	110–118	22.9–24.4	61.6–65.4
82	78.9–85.3	387.6–412.1	378.8–402.6	5.41	1960–2072	165–236	110–118	23.3–24.7	61.6–65.4
83	78.5–84.9	393.1–418.0	384.0–408.3	5.55	1961–2073	165–236	110–118	23.6–25.1	61.6–65.5
84	78.0–84.5	398.6–423.9	389.1–413.8	5.72	1961–2073	165–236	110–118	23.9–25.4	61.7–65.5
85	**77.6–84.1**	**404.0–429.8**	**394.3–419.4**	**5.87**	**1962–2074**	**165–236**	**110–118**	**24.2–25.8**	**61.7–65.5**
86	77.2–83.7	409.4–435.7	399.3–424.9	6.02	1962–2074	165–236	110–118	24.6–26.1	61.7–65.6
87	76.7–83.3	414.8–441.5	404.4–430.4	6.18	1962–2075	165–236	110–118	24.9–26.5	61.8–65.6
88	76.3–82.8	420.1–447.3	409.4–435.8	6.34	1963–2075	165–236	110–118	25.2–26.8	61.8–65.6
89	75.9–82.4	425.5–453.1	414.3–441.2	6.51	1963–2075	165–236	110–118	25.5–27.2	61.8–65.7
90	**75.4–82.0**	**430.7–458.8**	**419.3–446.5**	**6.69**	**1963–2076**	**165–236**	**110–118**	**25.8–27.5**	**61.9–65.7**
91	75.0–81.6	436.0–464.5	424.2–451.8	6.86	1964–2076	165–236	110–118	26.2–27.9	61.9–65.7
92	74.5–81.1	441.2–470.2	429.0–457.1	7.04	1964–2076	165–236	110–118	26.5–28.2	61.9–65.8
93	74.1–80.7	446.4–475.9	433.8–462.4	7.22	1964–2076	165–236	110–118	26.8–28.5	62.0–65.8
94	73.6–80.2	451.5–481.5	438.6–467.6	7.40	1964–2077	165–236	110–118	27.1–28.9	62.0–65.9
95	**73.1–79.8**	**456.7–487.1**	**443.3–472.7**	**7.60**	**1965–2077**	**165–236**	**110–118**	**27.4–29.2**	**62.0–65.9**
96	72.7–79.4	461.7–492.6	448.0–477.9	7.79	1965–2077	165–236	110–118	27.7–29.5	62.1–65.9
97	72.2–78.9	466.8–498.1	452.7–482.9	7.99	1965–2077	165–236	110–118	28.0–29.9	62.1–66.0
98	71.8–78.5	471.8–503.6	457.3–488.0	8.19	1965–2077	165–236	110–118	28.3–30.2	62.1–66.0
99	71.3–78.0	476.8–509.1	461.8–493.0	8.39	1965–2077	165–236	110–118	28.6–30.5	62.2–66.0
100	**70.8–77.6**	**481.8–514.5**	**466.4–497.9**	**8.60**	**1965–2078**	**165–236**	**110–118**	**28.9–30.8**	**62.2–66.1**

Production Period Space Recommendations

check local regulations concerning space requirements)

WEEKS OF AGE	4	15	20 30 40 50 60 70 80
CONVENTIONAL AND COLONY CAGES			
Floor Space			
	100–200 cm^2 (50–100 birds / m^2)	360 cm^2 (28 birds / m^2)	490 cm^2 (20 birds / m^2)
Nipple/Cup			
	1 / 12 birds	1 / 8 birds	1 / 12 birds or access to 2 drinkers
Feeders			
	4.5 cm / bird	5–7 cm / bird	7–9 cm / bird

Performance Graph



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Egg Quality and Egg Size Distribution

AGE (weeks)	HAUGH UNITS	BREAKING STRENGTH	SHELL COLOR
20	**97.8**	**4605**	**89**
22	97.0	4590	89
24	96.0	4580	89
26	95.1	4570	88
28	94.2	4560	88
30	**93.3**	**4540**	**88**
32	92.2	4515	88
34	91.5	4490	88
36	90.6	4450	87
38	90.0	4425	87
40	**89.3**	**4405**	**87**
42	88.5	4375	87
44	87.8	4355	87
46	87.1	4320	87
48	86.4	4305	87
50	**85.6**	**4280**	**86**
52	85.0	4250	86
54	84.6	4225	86
56	84.0	4190	85
58	83.1	4170	85
60	**82.6**	**4150**	**85**
62	82.2	4130	84
64	81.9	4110	83
66	81.6	4095	83
68	81.5	4085	82
70	**81.1**	**4075**	**81**
72	81.0	4065	81
74	80.8	4055	80
76	80.5	4040	80
78	80.2	4020	80
80	**80.1**	**3995**	**80**
82	80.0	3985	79
84	79.9	3975	79
86	79.8	3965	79
88	79.7	3960	79
90	**79.7**	**3955**	**79**

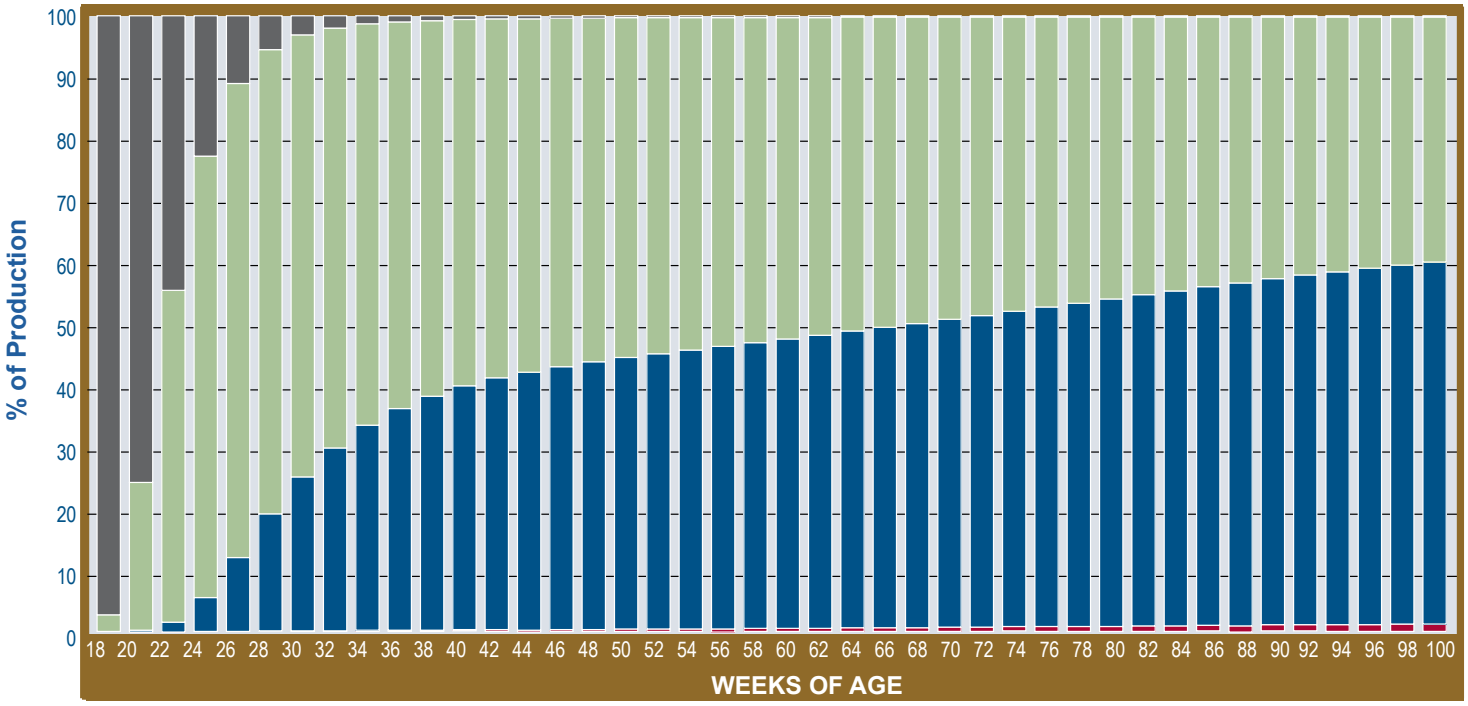
E.U. Standards–Weekly*

AGE (weeks)	AVERAGE EGG WEIGHT (g)	WEEKLY % SMALL 43–53 g	WEEKLY % MEDIUM 53–63 g	WEEKLY % LARGE 63–73 g	WEEKLY % VERY LARGE Over 73 g
18	44.1	97.3	2.7	0.0	0.0
20	**49.0**	**75.8**	**24.0**	**0.3**	**0.0**
22	52.1	44.6	53.9	1.6	0.0
24	54.8	22.8	71.7	5.5	0.0
26	56.9	11.0	77.0	12.0	0.0
28	58.6	5.5	75.4	19.0	0.0
30	**59.8**	**3.1**	**71.8**	**25.0**	**0.1**
32	60.7	2.0	68.2	29.7	0.1
34	61.4	1.3	65.2	33.3	0.2
36	61.8	1.0	62.8	36.0	0.2
38	62.0	0.8	61.0	38.0	0.2
40	**62.1**	**0.6**	**59.5**	**39.6**	**0.2**
42	62.1	0.5	58.3	40.9	0.3
44	62.2	0.5	57.4	41.9	0.3
46	62.3	0.4	56.6	42.7	0.3
48	62.3	0.4	55.8	43.5	0.3
50	**62.4**	**0.3**	**55.2**	**44.1**	**0.4**
52	62.5	0.3	54.6	44.7	0.4
54	62.6	0.3	54.0	45.3	0.4
56	62.6	0.3	53.4	45.9	0.5
58	62.7	0.3	52.8	46.4	0.5
60	**62.8**	**0.3**	**52.2**	**47.0**	**0.5**
62	62.8	0.3	51.6	47.6	0.5
64	62.9	0.2	51.0	48.2	0.6
66	63.0	0.2	50.4	48.8	0.6
68	63.0	0.2	49.8	49.4	0.6
70	**63.1**	**0.2**	**49.1**	**50.0**	**0.7**
72	63.2	0.2	48.5	50.6	0.7
74	63.2	0.2	47.8	51.2	0.7
76	63.3	0.2	47.1	51.9	0.8
78	63.4	0.2	46.5	52.5	0.8
80	**63.5**	**0.2**	**45.8**	**53.2**	**0.8**
82	63.5	0.2	45.1	53.8	0.9
84	63.6	0.2	44.5	54.4	0.9
86	63.7	0.2	43.8	55.0	0.9
88	63.7	0.2	43.2	55.7	1.0
90	**63.8**	**0.2**	**42.5**	**56.2**	**1.0**
92	63.9	0.2	41.9	56.8	1.0
94	63.9	0.2	41.4	57.3	1.1
96	64.0	0.2	40.8	57.9	1.1
98	64.1	0.2	40.3	58.3	1.2
100	**64.1**	**0.2**	**39.8**	**58.8**	**1.2**

** Distribution of egg sizes based on weekly (not cumulative) average egg weights.*

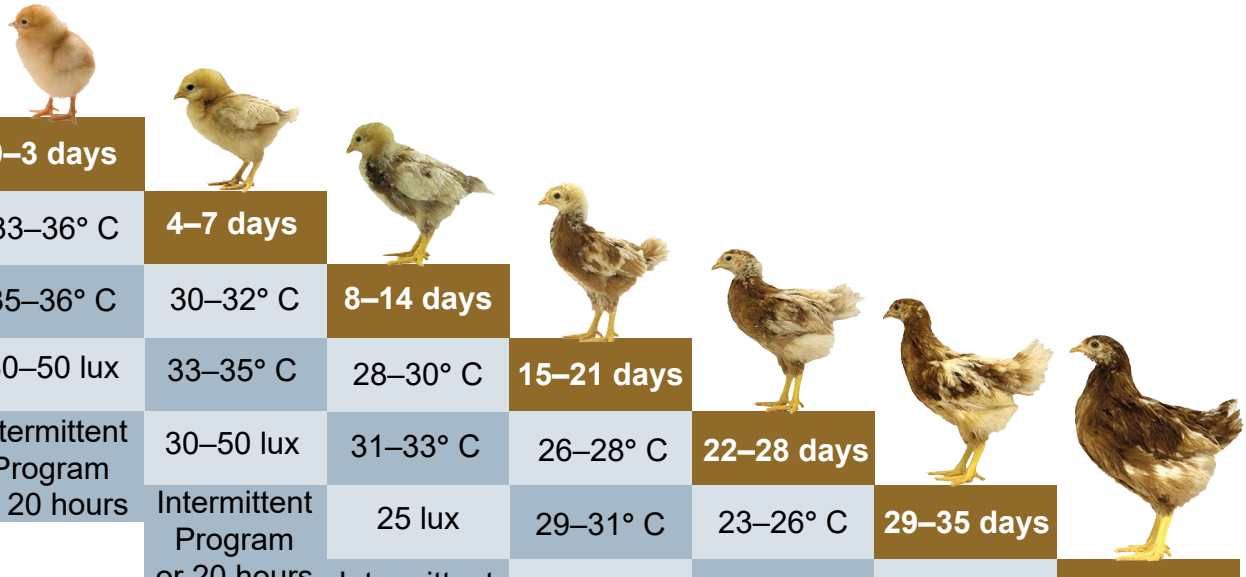
Egg Size Distribution (cont.)

E.U. Standards–Weekly*



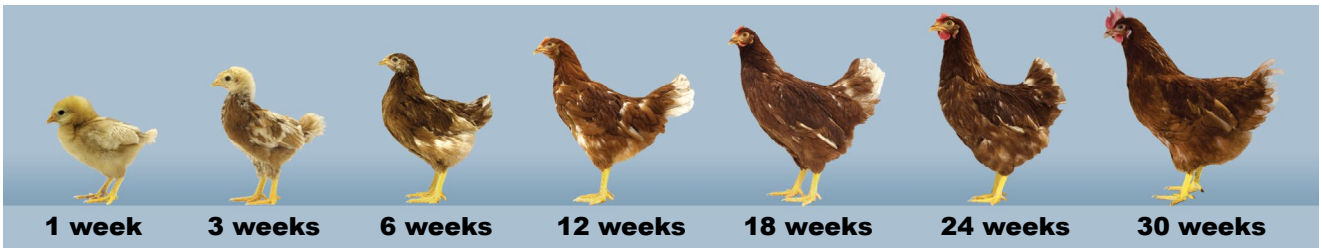
** Distribution of egg sizes based on weekly (not cumulative) average egg weights.*

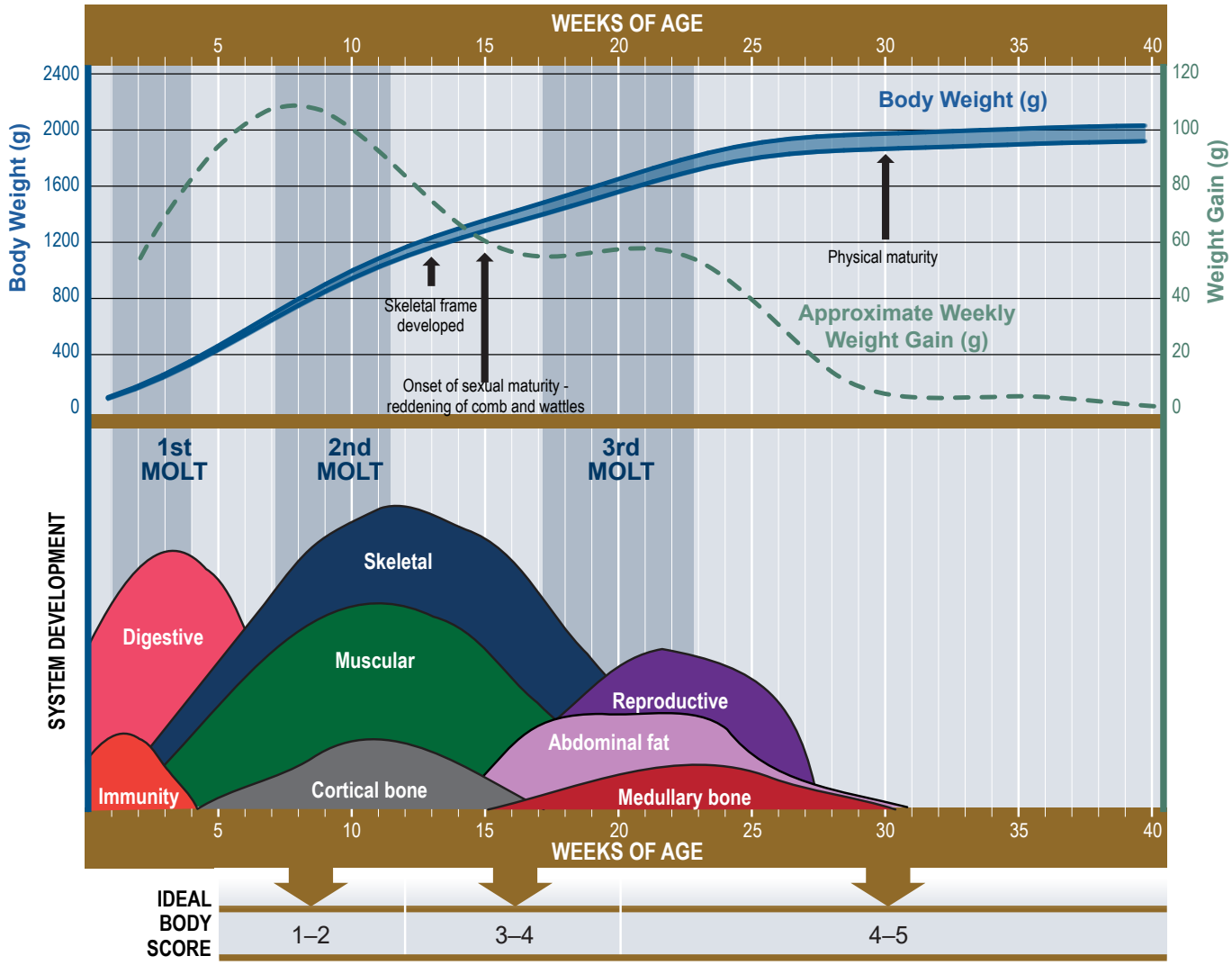
Brooding Temperature and Lighting Recommendations



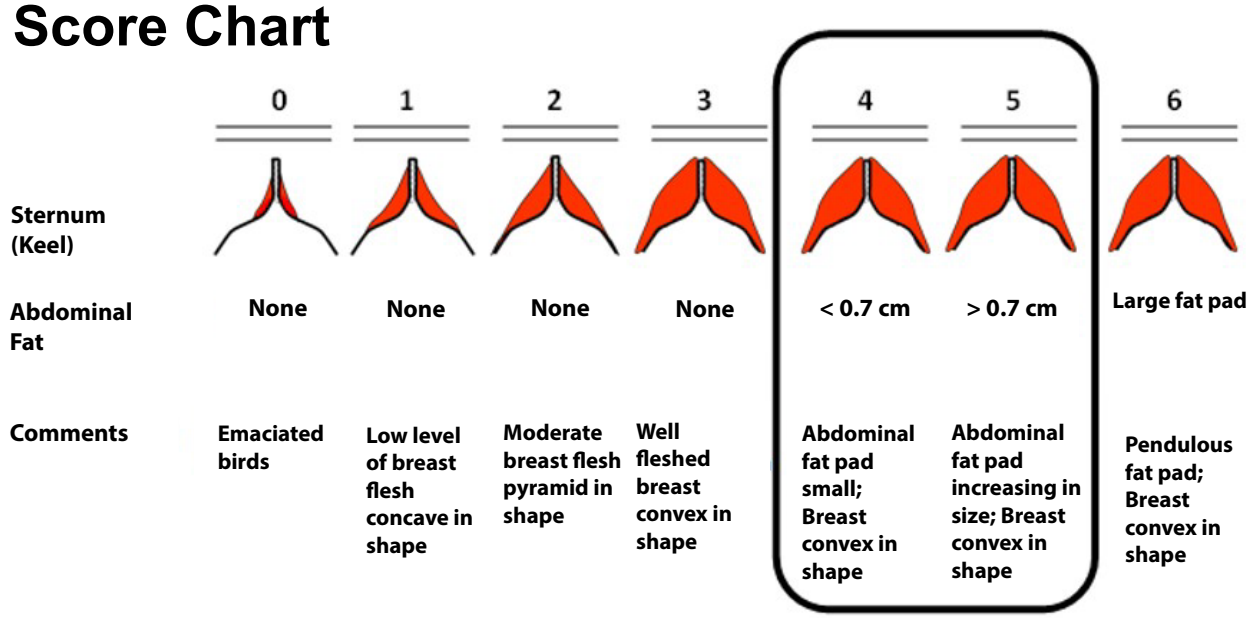
AGE	0–3 days	4–7 days	8–14 days	15–21 days	22–28 days	29–35 days	36–42 days
AIR TEMP. (CAGE)	33–36° C	30–32° C	28–30° C	26–28° C	23–26° C	21–23° C	21° C
AIR TEMP. (FLOOR)	35–36° C	33–35° C	31–33° C	29–31° C	26–27° C	23–25° C	21° C
LIGHT INTENSITY	30–50 lux	30–50 lux	25 lux	25 lux	25 lux	10 lux	10 lux
LIGHT HOURS	Intermittent Program or 20 hours	Intermittent Program or 20 hours	Intermittent Program or 20 hours	18 hours	16 hours	14 hours	12 hours

Development of the Organ Systems in Pullets



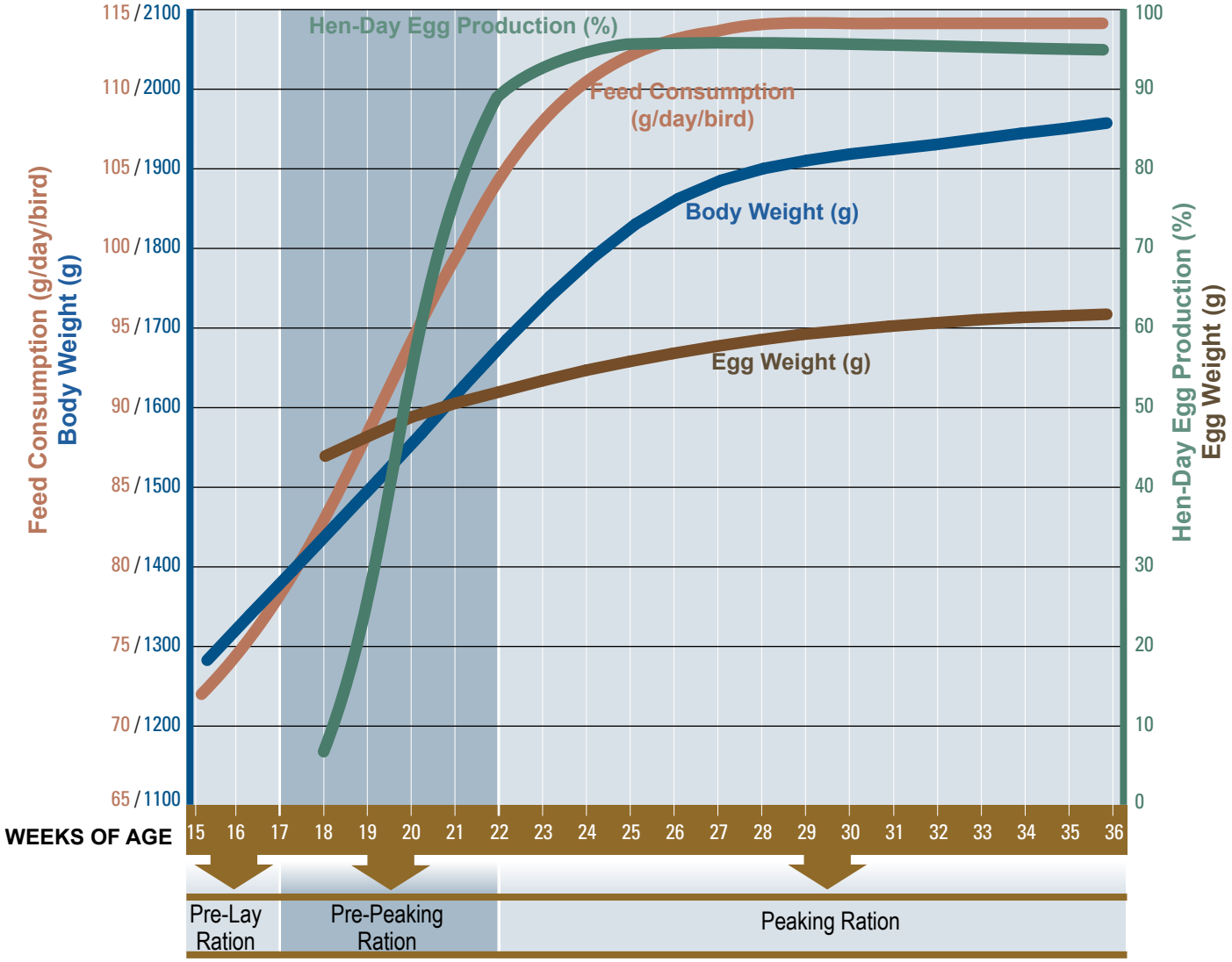


Body Score Chart



Transition Period from Rear to Peak Egg Production

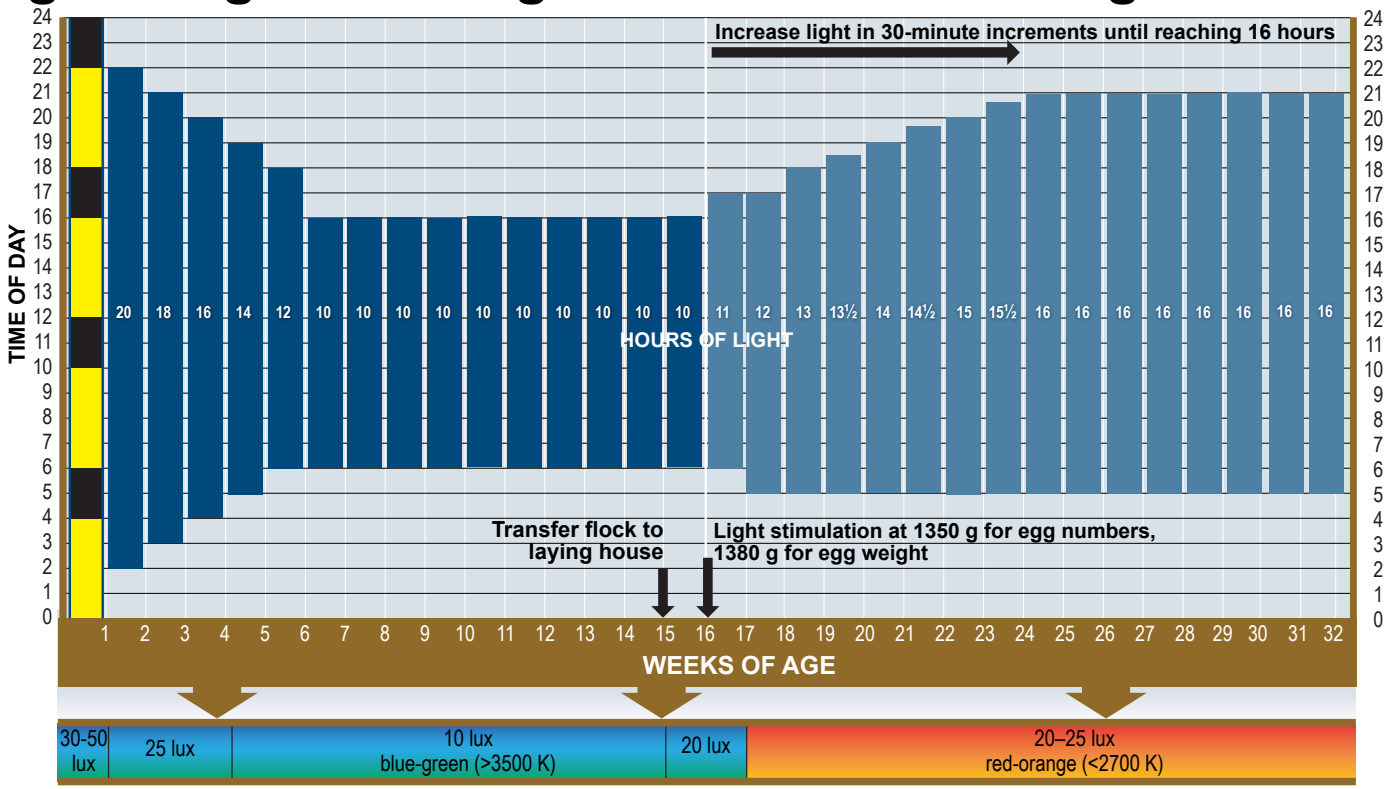
Frequently formulate to changing feed consumption during transition period until feed consumption is consistent.



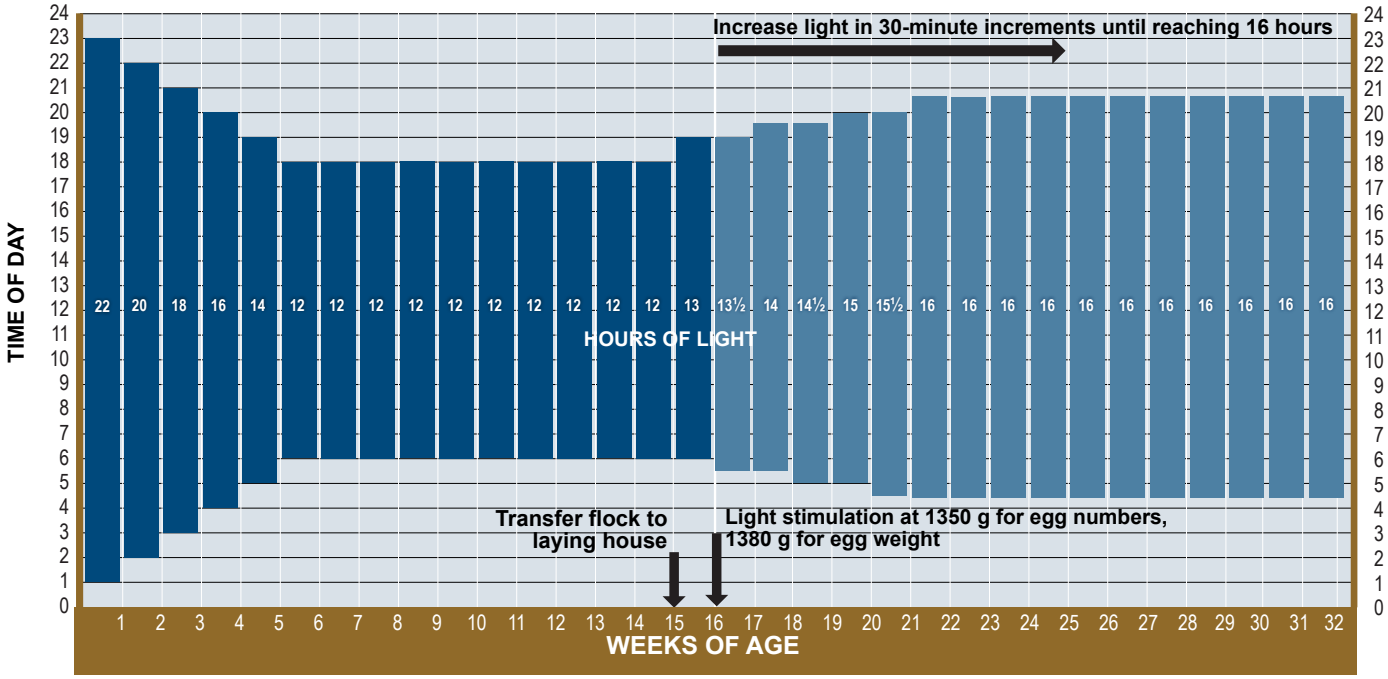
Pre-Peak

- Pre-Peak diets are intended for flocks with low feed intake and are fed for a limited period from first egg to the beginning of peak production. The nutrient specification of the Pre-Peak diet should be dense enough to allow for lower feed intake and also cater to the increased nutritional needs of the bird entering egg production. Continue to feed the Pre-Peak until feed intake has developed sufficiently to allow transition to the Peak diet.
- If utilized until no more than 50–70% HD, a Pre-Peak diet with reduced energy concentration can be beneficial to stimulate feed intake. Pre-Peaking diets are useful in situations where local conditions may result in reduced feed intake, such as hot climates where feed intake may be depressed.
- Increasing the vitamins and trace mineral inclusion to 30% can be useful to cope with the lower feed intake during the Pre-Peak phase.

Light Program for Light-Controlled Housing

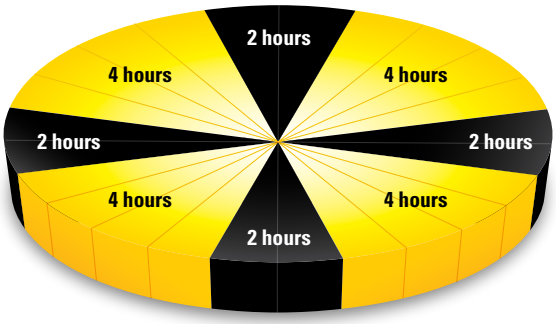


Light Program for Open-Sided Housing

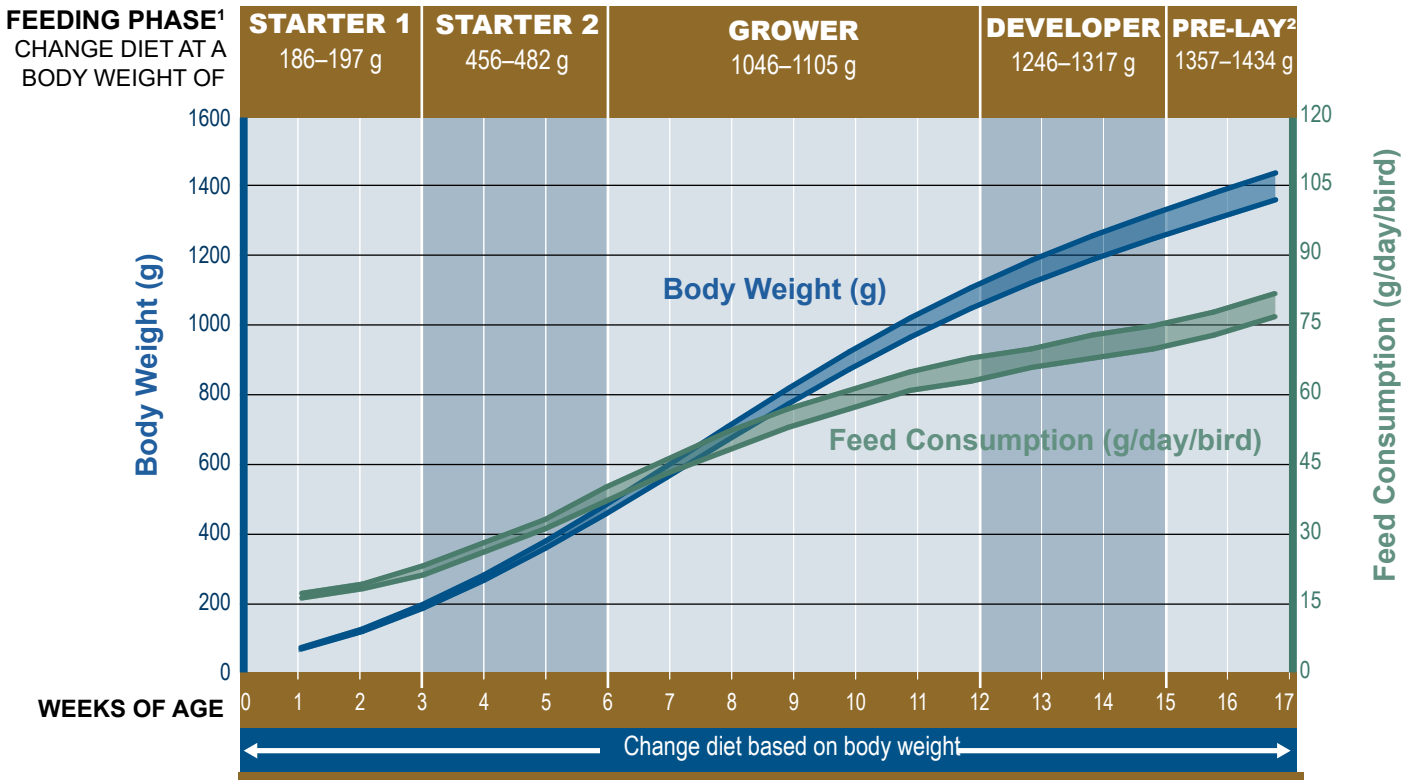


Intermittent Lighting Program for Chicks

- Preferred lighting technique.
- Use from 0–7 days (can be used up to 14 days of age).
- Intermittent dark periods provide rest periods for chicks.
- Synchronises chicks' activities and feedings.
- Establishes more natural behaviour of rest and activity.
- May improve 7-day livability and pullet body weight.
- Some dark periods may be shortened or removed to accommodate work schedules.



Rearing Period Nutritional Recommendations



NUTRITION	RECOMMENDED NUTRIENT CONCENTRATION				
Metabolizable energy[3], kcal/kg	2900 / 3050	2850 / 3000	2700 / 2850	2600 / 2750	2650 / 2800
Metabolizable energy[3], MJ/kg	12.13 / 12.76	11.92 / 12.55	11.30 / 11.92	10.88 / 11.51	11.09 / 11.72
	Standardized Ileal Digestible Amino Acids / Total Amino Acids[4]				
Lysine, %	1.00 / 1.10	0.87 / 0.95	0.76 / 0.83	0.62 / 0.68	0.75 / 0.82
Methionine, %	0.50 / 0.54	0.43 / 0.46	0.36 / 0.40	0.29 / 0.31	0.38 / 0.40
Methionine+Cystine,%	0.90 / 1.01	0.77 / 0.86	0.65 / 0.73	0.52 / 0.59	0.68 / 0.76
Threonine, %	0.71 / 0.84	0.61 / 0.71	0.52 / 0.61	0.42 / 0.50	0.54 / 0.63
Tryptophan, %	0.20 / 0.23	0.17 / 0.20	0.15 / 0.18	0.12 / 0.15	0.17 / 0.21
Arginine, %	1.04 / 1.12	0.90 / 0.97	0.79 / 0.85	0.64 / 0.69	0.78 / 0.85
Isoleucine, %	0.76 / 0.81	0.66 / 0.71	0.57 / 0.61	0.46 / 0.49	0.59 / 0.64
Valine, %	0.87 / 0.95	0.75 / 0.82	0.65 / 0.71	0.52 / 0.57	0.65 / 0.72
Crude protein[5], %	18.52	16.73	15.51	15.00	16.30
Calcium[6], %	1.05	1.00	0.95	0.90	2.50
Phosphorus (available)[7], %	0.45	0.44	0.43	0.38	0.42
Phosphorus (digestible), %	0.41	0.40	0.39	0.34	0.38
Sodium, %	0.18	0.18	0.18	0.18	0.18
Chloride, %	0.18	0.18	0.18	0.18	0.18
Linoleic acid (C18:2 n-6), %	1.20	1.20	1.20	1.20	1.20
Choline, mg/kg	2,000	1,800	1,800	1,500	1,800

[1] Body weights are approximate. Ages shown are a guide only. Please note that at time of transfer, there will be some loss in body weight (normally 10–12%) due to reduced water intake.

[2] Do not feed Pre-Lay Diet earlier than 15 weeks of age. Do not feed Pre-Lay later than first egg as it contains insufficient calcium to support egg production. Implementing a pre-lay diet can be challenging in mixed-age flocks. If it's not possible to use the Pre-Lay diet, the calcium content of the last stage rearing diet (developer) must be increased to 1.4%.

[3] Recommended energy range is based on raw material energy values shown in feed ingredient table at back of this guide. It is important that target concentrations of dietary energy are adjusted according to energy system applied to raw material matrix.

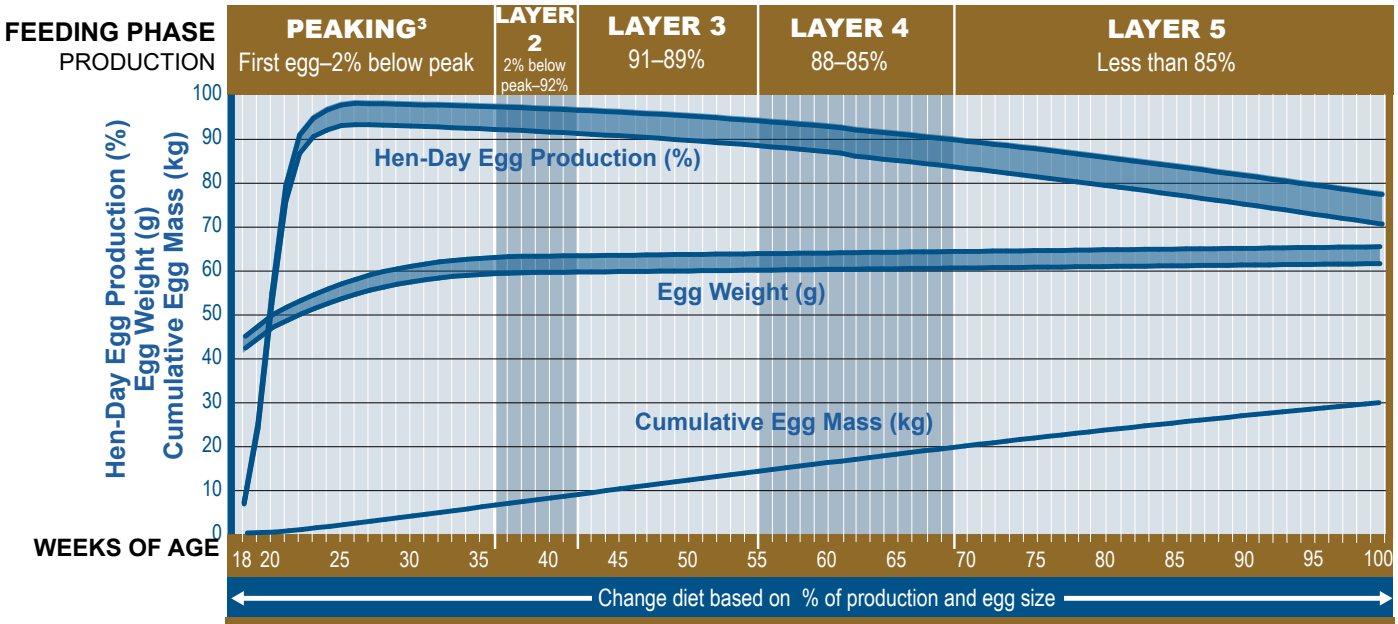
[4] Recommendation for Total Amino Acids is only appropriate to corn and soybean meal diet. Where diets utilize other ingredients, recommendations for Standardized Ileal Digestible Amino Acids must be followed.

[5] Diets should always be formulated to provide required intake of amino acid. Concentration of crude protein in diet will vary with raw materials used. Crude protein value provided is an estimated typical value only.

[6] Calcium should be supplied as fine calcium carbonate (mean particle size less than 2 mm). Coarse limestone (2–4 mm) can be introduced in Pre-Lay Diet at up to 50% of total limestone.

[7] Where other phosphorus systems are used, diets should contain recommended minimum level of available phosphorus.

Production Period Nutritional Recommendations for Egg Numbers[1,2]



NUTRITION	RECOMMENDED DAILY NUTRIENT INTAKE				
Metabolizable energy[4], kcal/bird/day	320 / 330	320 / 330	317 / 327	314 / 324	311 / 321
Metabolizable energy[4], MJ/bird/day	1.34 / 1.38	1.34 / 1.38	1.33 / 1.37	1.31 / 1.36	1.30 / 1.34
	Standardized Ileal Digestible Amino Acids / Total Amino Acids[5]				
Lysine, mg/day	830 / 909	830 / 909	800 / 876	770 / 843	740 / 810
Methionine , mg/day	415 / 446	415 / 446	400 / 430	385 / 414	370 / 398
Methionine+Cystine, mg/day	747 / 842	747 / 842	720 / 812	693 / 782	666 / 751
Threonine, mg/day	606 / 713	606 / 713	584 / 687	562 / 661	540 / 636
Tryptophan, mg/day	183 / 218	183 / 218	176 / 210	169 / 202	163 / 195
Arginine, mg/day	863 / 928	863 / 928	832 / 894	801 / 861	770 / 828
Isoleucine, mg/day	664 / 714	664 / 714	640 / 688	616 / 662	592 / 637
Valine, mg/day	730 / 806	730 / 806	704 / 776	678 / 747	651 / 718
Crude protein[6], g/day	17.85	17.85	17.20	16.50	15.90
Sodium, mg/day	180	170	170	170	170
Chloride, mg/day	180	170	170	170	170
Linoleic acid (C18:2 n-6), g/day	1.60	1.50	1.40	1.40	1.40
Choline, mg/day	180	180	180	180	180

	CALCIUM AND PHOSPHORUS			
	Calcium[7,8] g/day	Phosphorus (available)[7,9] mg/day	Phosphorus (digestible) mg/day	Calcium Particle Size (fine : coarse)
Weeks 18–33	4.00	432	389	40% : 60%
Weeks 34–48	4.20	405	366	35% : 65%
Weeks 49–62	4.40	373	337	30% : 70%
Weeks 63–76	4.60	347	314	25% : 75%
Weeks 77+	4.70	324	291	25% : 75%

	IDEAL PROTEIN REFERENCE				
	PEAKING	LAYER 2	LAYER 3	LAYER 4	LAYER 5
Lysine	100%	100%	100%	100%	100%
Methionine	50%	50%	50%	50%	50%
M+C	90%	90%	90%	90%	90%
Threonine	73%	73%	73%	73%	73%
Tryptophan	22%	22%	22%	22%	22%
Arginine	104%	104%	104%	104%	104%
Isoleucine	80%	80%	80%	80%	80%
Valine	88%	88%	88%	88%	88%

Production Period Dietary Nutrient Concentrations for Egg Numbers[1,2]

FEEDING PHASE	PEAKING[3] First egg until production drops 2% below peak					LAYER 2 2% below peak to 92%					LAYER 3 91–89%					LAYER 4 88–85%					LAYER 5 Less than 85%				
PRODUCTION NUTRITION	**RECOMMENDED CONCENTRATION**																								
Metabolizable energy[4], kcal/bird/day	320 / 330					320 / 330					317 / 327					314 / 324					311 / 321				
Metabolizable energy[4], MJ/bird/day	1.34 / 1.38					1.34 / 1.38					1.33 / 1.37					1.31 / 1.36					1.30 / 1.34				
FEED CONSUMPTION (*Typical Feed Consumption)																									
g/day per bird	90	95	**100***	105	110	100	105	**110***	115	120	100	105	**110***	115	120	100	105	**110***	115	120	100	105	**110***	115	120
Standardized Ileal Digestible Amino Acids																									
Lysine, %	0.92	0.87	**0.83**	0.79	0.75	0.83	0.79	**0.75**	0.72	0.69	0.80	0.76	**0.73**	0.70	0.67	0.77	0.73	**0.70**	0.67	0.64	0.74	0.70	**0.67**	0.64	0.62
Methionine, %	0.46	0.44	**0.42**	0.40	0.38	0.42	0.40	**0.38**	0.36	0.35	0.40	0.38	**0.36**	0.35	0.33	0.39	0.37	**0.35**	0.33	0.32	0.37	0.35	**0.34**	0.32	0.31
Methionine+Cystine, %	0.83	0.79	**0.75**	0.71	0.68	0.75	0.71	**0.68**	0.65	0.62	0.72	0.69	**0.65**	0.63	0.60	0.69	0.66	**0.63**	0.60	0.58	0.67	0.63	**0.61**	0.58	0.56
Threonine, %	0.67	0.64	**0.61**	0.58	0.55	0.61	0.58	**0.55**	0.53	0.51	0.58	0.56	**0.53**	0.51	0.49	0.56	0.54	**0.51**	0.49	0.47	0.54	0.51	**0.49**	0.47	0.45
Tryptophan, %	0.20	0.19	**0.18**	0.17	0.17	0.18	0.17	**0.17**	0.16	0.15	0.18	0.17	**0.16**	0.15	0.15	0.17	0.16	**0.15**	0.15	0.14	0.16	0.16	**0.15**	0.14	0.14
Arginine, %	0.96	0.91	**0.86**	0.82	0.78	0.86	0.82	**0.78**	0.75	0.72	0.83	0.79	**0.76**	0.72	0.69	0.80	0.76	**0.73**	0.70	0.67	0.77	0.73	**0.70**	0.67	0.64
Isoleucine, %	0.74	0.70	**0.66**	0.63	0.60	0.66	0.63	**0.60**	0.58	0.55	0.64	0.61	**0.58**	0.56	0.53	0.62	0.59	**0.56**	0.54	0.51	0.59	0.56	**0.54**	0.51	0.49
Valine, %	0.81	0.77	**0.73**	0.70	0.66	0.73	0.70	**0.66**	0.63	0.61	0.70	0.67	**0.64**	0.61	0.59	0.68	0.65	**0.62**	0.59	0.57	0.65	0.62	**0.59**	0.57	0.54
Total Amino Acids[5]																									
Lysine, %	1.01	0.96	**0.91**	0.87	0.83	0.91	0.87	**0.83**	0.79	0.76	0.88	0.83	**0.80**	0.76	0.73	0.84	0.80	**0.77**	0.73	0.70	0.81	0.77	**0.74**	0.70	0.68
Methionine, %	0.50	0.47	**0.45**	0.42	0.41	0.45	0.42	**0.41**	0.39	0.37	0.43	0.41	**0.39**	0.37	0.36	0.41	0.39	**0.38**	0.36	0.35	0.40	0.38	**0.36**	0.35	0.33
Methionine+Cystine, %	0.94	0.89	**0.84**	0.80	0.77	0.84	0.80	**0.77**	0.73	0.70	0.81	0.77	**0.74**	0.71	0.68	0.78	0.74	**0.71**	0.68	0.65	0.75	0.72	**0.68**	0.65	0.63
Threonine, %	0.79	0.75	**0.71**	0.68	0.65	0.71	0.68	**0.65**	0.62	0.59	0.69	0.65	**0.62**	0.60	0.57	0.66	0.63	**0.60**	0.57	0.55	0.64	0.61	**0.58**	0.55	0.53
Tryptophan, %	0.24	0.23	**0.22**	0.21	0.20	0.22	0.21	**0.20**	0.19	0.18	0.21	0.20	**0.19**	0.18	0.18	0.20	0.19	**0.18**	0.18	0.17	0.20	0.19	**0.18**	0.17	0.16
Arginine, %	1.03	0.98	**0.93**	0.88	0.84	0.93	0.88	**0.84**	0.81	0.77	0.89	0.85	**0.81**	0.78	0.75	0.86	0.82	**0.78**	0.75	0.72	0.83	0.79	**0.75**	0.72	0.69
Isoleucine, %	0.79	0.75	**0.71**	0.68	0.65	0.71	0.68	**0.65**	0.62	0.60	0.69	0.66	**0.63**	0.60	0.57	0.66	0.63	**0.60**	0.58	0.55	0.64	0.61	**0.58**	0.55	0.53
Valine, %	0.90	0.85	**0.81**	0.77	0.73	0.81	0.77	**0.73**	0.70	0.67	0.78	0.74	**0.71**	0.67	0.65	0.75	0.71	**0.68**	0.65	0.62	0.72	0.68	**0.65**	0.62	0.60
Crude protein[6], %	19.83	18.79	**17.85**	17.00	16.23	17.85	17.00	**16.23**	15.52	14.88	17.20	16.38	**15.64**	14.96	14.33	16.50	15.71	**15.00**	14.35	13.75	15.90	15.14	**14.45**	13.83	13.25
Sodium, %	0.20	0.19	**0.18**	0.17	0.16	0.17	0.16	**0.15**	0.15	0.14	0.17	0.16	**0.15**	0.15	0.14	0.17	0.16	**0.15**	0.15	0.14	0.17	0.16	**0.15**	0.15	0.14
Chloride, %	0.20	0.19	**0.18**	0.17	0.16	0.17	0.16	**0.15**	0.15	0.14	0.17	0.16	**0.15**	0.15	0.14	0.17	0.16	**0.15**	0.15	0.14	0.17	0.16	**0.15**	0.15	0.14
Linoleic acid (C18:2 n-6), %	1.78	1.68	**1.60**	1.52	1.45	1.50	1.43	**1.36**	1.30	1.25	1.40	1.33	**1.27**	1.22	1.17	1.40	1.33	**1.27**	1.22	1.17	1.40	1.33	**1.27**	1.22	1.17
Choline, mg/kg	2000	1895	**1800**	1714	1636	1800	1714	**1636**	1565	1500	1800	1714	**1636**	1565	1500	1800	1714	**1636**	1565	1500	1800	1714	**1636**	1565	1500

CALCIUM AND PHOSPHORUS CHANGES BASED ON FEED INTAKE																									
	Weeks 18–33							Weeks 34–48					Weeks 49–62					Weeks 63–76					Weeks 77+				
Feed Consumption, g/day per bird	90	95	100	**105**	110	115	120	100	105	**110**	115	120	100	105	**110**	115	120	100	105	**110**	115	120	100	105	**110**	115	120
Calcium[7,8], %	4.44	4.21	4.00	**3.81**	3.64	3.48	3.33	4.20	4.00	**3.82**	3.65	3.50	4.40	4.19	**4.00**	3.83	3.67	4.60	4.38	**4.18**	4.00	3.83	4.70	4.48	**4.27**	4.09	3.92
Phosphorus (available)[7,9], %	0.48	0.46	0.43	**0.41**	0.39	0.38	0.36	0.41	0.39	**0.37**	0.35	0.34	0.37	0.36	**0.34**	0.32	0.31	0.35	0.33	**0.32**	0.30	0.29	0.32	0.31	**0.29**	0.28	0.27
Phosphorus (digestible), %	0.43	0.41	0.39	**0.37**	0.35	0.34	0.32	0.37	0.35	**0.33**	0.32	0.31	0.34	0.32	**0.31**	0.29	0.28	0.31	0.30	**0.29**	0.27	0.26	0.29	0.28	**0.26**	0.25	0.24

[1] All nutrient requirements are based on the Feed Ingredient Tables.

[2] Crude protein, methionine+cystine, fat, linoleic acid, and / or energy may be changed to optimize egg size.

[3] Peaking nutrient levels are calculated for birds at peak egg production. Prior to achieving peak egg production, the nutrient requirements will be lower.

[4] A good approximation of the influence of temperature on energy needs is that for each 0.5°C change higher or lower than 22°C, subtract or add about 2 kcal /bird /day, respectively.

[5] Recommendation for Total Amino Acids is only appropriate to corn and soybean meal diet. Where diets utilize other ingredients, recommendations for Standardized Ileal Digestible Amino Acids must be followed.

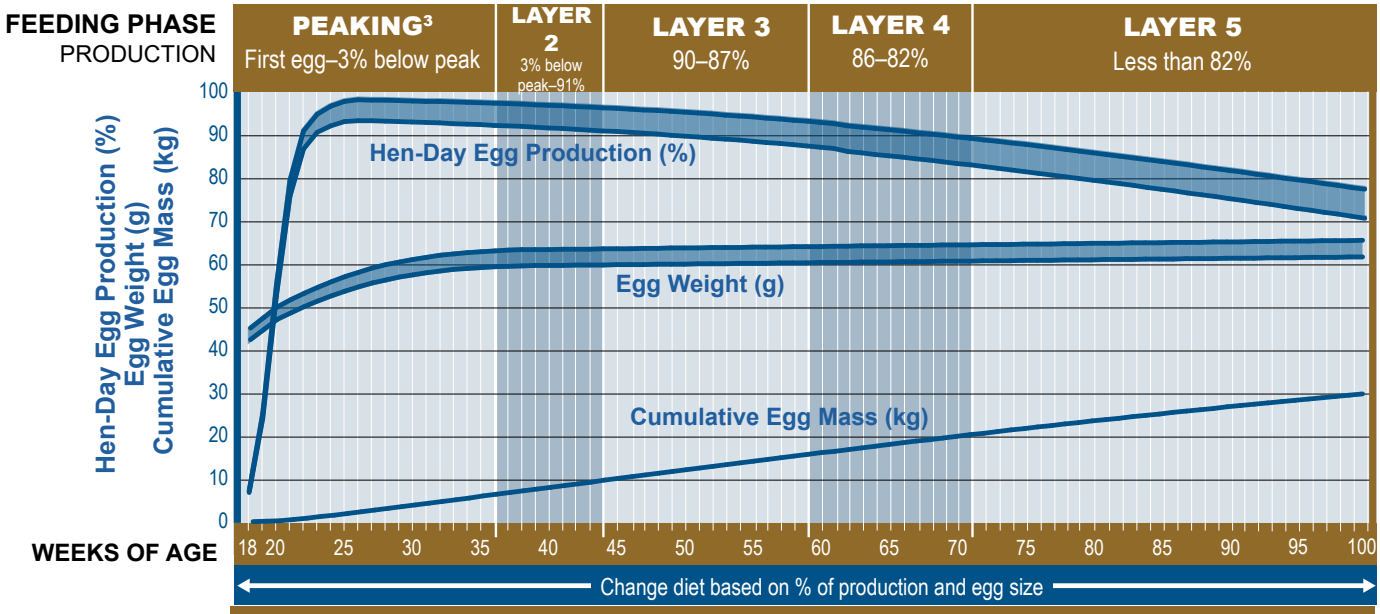
[6] Diets should always be formulated to provide required intake of amino acid. Concentration of crude protein in diet will vary with raw material used. Crude protein value provided is an estimated typical value only.

[7] Calcium and available phosphorus requirements are determined by flock age. When production remains higher and diets are fed for longer than ages shown, it is recommended to increase to calcium and phosphorus concentrations of next feeding phase.

[8] Calcium carbonate particle size recommendation varies throughout lay. Refer to Calcium Particle Size. Dietary calcium levels and fine to coarse ratio may need to be adjusted based on limestone solubility.

[9] Where other phosphorus systems are used, diets should contain recommended minimum level of available phosphorus.

Production Period Nutritional Recommendations for Egg Weight[1,2]



NUTRITION	RECOMMENDED DAILY NUTRIENT INTAKE				
Metabolizable energy[4], kcal/bird/day	320 / 330	320 / 330	317 / 327	314 / 324	311 / 321
Metabolizable energy[4], MJ/bird/day	1.34 / 1.38	1.34 / 1.38	1.33 / 1.37	1.31 / 1.36	1.30 / 1.34
	Standardized Ileal Digestible Amino Acids / Total Amino Acids[5]				
Lysine, mg/day	870 / 953	870 / 953	830 / 909	800 / 876	770 / 843
Methionine , mg/day	435 / 468	435 / 468	415 / 446	400 / 430	385 / 414
Methionine+Cystine, mg/day	800 / 903	792 / 893	747 / 842	720 / 812	693 / 782
Threonine, mg/day	635 / 747	635 / 747	606 / 713	584 / 687	562 / 661
Tryptophan, mg/day	191 / 229	191 / 229	183 / 218	176 / 210	169 / 202
Arginine, mg/day	905 / 973	905 / 973	863 / 928	832 / 895	801 / 861
Isoleucine, mg/day	713 / 767	705 / 758	664 / 714	640 / 688	616 / 662
Valine, mg/day	783 / 864	774 / 854	730 / 806	704 / 776	678 / 747
Crude protein[6], g/day	18.70	18.70	17.80	17.20	16.50
Sodium, mg/day	180	170	170	170	170
Chloride, mg/day	180	170	170	170	170
Linoleic acid (C18:2 n-6), g/day	2.00	2.00	1.60	1.50	1.40
Choline, mg/day	160	180	180	180	180

	CALCIUM AND PHOSPHORUS			
	Calcium[7,8] g/day	Phosphorus (available)[7,9] mg/day	Phosphorus (digestible) mg/day	Calcium Particle Size (fine : coarse)
Weeks 18–33	4.00	432	389	40% : 60%
Weeks 34–48	4.20	405	366	35% : 65%
Weeks 49–62	4.40	373	337	30% : 70%
Weeks 63–76	4.60	347	314	25% : 75%
Weeks 77+	4.70	324	291	25% : 75%

	IDEAL PROTEIN REFERENCE				
	PEAKING	LAYER 2	LAYER 3	LAYER 4	LAYER 5
Lysine	100%	100%	100%	100%	100%
Methionine	50%	50%	50%	50%	50%
M+C	92%	91%	90%	90%	90%
Threonine	73%	73%	73%	73%	73%
Tryptophan	22%	22%	22%	22%	22%
Arginine	104%	104%	104%	104%	104%
Isoleucine	82%	81%	80%	80%	80%
Valine	90%	89%	88%	88%	88%

Production Period Dietary Nutrient Concentrations for Egg Weight[1,2]

FEEDING PHASE	PEAKING[3] First egg until production drops 3% below peak					LAYER 2 3% below peak to 91%					LAYER 3 90–87%					LAYER 4 86–82%					LAYER 5 Less than 82%				
PRODUCTION NUTRITION	**RECOMMENDED CONCENTRATION**																								
Metabolizable energy[4], kcal/bird/day	320 / 330					320 / 330					317 / 327					314 / 324					311 / 321				
Metabolizable energy[4], MJ/bird/day	1.34 / 1.38					1.34 / 1.38					1.33 / 1.37					1.31 / 1.36					1.30 / 1.34				
FEED CONSUMPTION (*Typical Feed Consumption)																									
g/day per bird	90	95	**100***	105	110	100	105	**110***	115	120	100	105	**110***	115	120	100	105	**110***	115	120	100	105	**110***	115	120
Standardized Ileal Digestible Amino Acids																									
Lysine, %	0.97	0.92	**0.87**	0.83	0.79	0.87	0.83	**0.79**	0.76	0.73	0.83	0.79	**0.75**	0.72	0.69	0.80	0.76	**0.73**	0.70	0.67	0.77	0.73	**0.70**	0.67	0.64
Methionine, %	0.48	0.46	**0.44**	0.41	0.40	0.44	0.41	**0.40**	0.38	0.36	0.42	0.40	**0.38**	0.36	0.35	0.40	0.38	**0.36**	0.35	0.33	0.39	0.37	**0.35**	0.33	0.32
Methionine+Cystine, %	0.89	0.84	**0.80**	0.76	0.73	0.79	0.75	**0.72**	0.69	0.66	0.75	0.71	**0.68**	0.65	0.62	0.72	0.69	**0.65**	0.63	0.60	0.69	0.66	**0.63**	0.60	0.58
Threonine, %	0.71	0.67	**0.64**	0.60	0.58	0.64	0.60	**0.58**	0.55	0.53	0.61	0.58	**0.55**	0.53	0.51	0.58	0.56	**0.53**	0.51	0.49	0.56	0.54	**0.51**	0.49	0.47
Tryptophan, %	0.21	0.20	**0.19**	0.18	0.17	0.19	0.18	**0.17**	0.17	0.16	0.18	0.17	**0.17**	0.16	0.15	0.18	0.17	**0.16**	0.15	0.15	0.17	0.16	**0.15**	0.15	0.14
Arginine, %	1.01	0.95	**0.91**	0.86	0.82	0.91	0.86	**0.82**	0.79	0.75	0.86	0.82	**0.78**	0.75	0.72	0.83	0.79	**0.76**	0.72	0.69	0.80	0.76	**0.73**	0.70	0.67
Isoleucine, %	0.79	0.75	**0.71**	0.68	0.65	0.71	0.67	**0.64**	0.61	0.59	0.66	0.63	**0.60**	0.58	0.55	0.64	0.61	**0.58**	0.56	0.53	0.62	0.59	**0.56**	0.54	0.51
Valine, %	0.87	0.82	**0.78**	0.75	0.71	0.77	0.74	**0.70**	0.67	0.65	0.73	0.70	**0.66**	0.63	0.61	0.70	0.67	**0.64**	0.61	0.59	0.68	0.65	**0.62**	0.59	0.57
Total Amino Acids[5]																									
Lysine, %	1.06	1.00	**0.95**	0.91	0.87	0.95	0.91	**0.87**	0.83	0.79	0.91	0.87	**0.83**	0.79	0.76	0.88	0.83	**0.80**	0.76	0.73	0.84	0.80	**0.77**	0.73	0.70
Methionine, %	0.52	0.49	**0.47**	0.45	0.43	0.47	0.45	**0.43**	0.41	0.39	0.45	0.42	**0.41**	0.39	0.37	0.43	0.41	**0.39**	0.37	0.36	0.41	0.39	**0.38**	0.36	0.35
Methionine+Cystine, %	1.00	0.95	**0.90**	0.86	0.82	0.89	0.85	**0.81**	0.78	0.74	0.84	0.80	**0.77**	0.73	0.70	0.81	0.77	**0.74**	0.71	0.68	0.78	0.74	**0.71**	0.68	0.65
Threonine, %	0.83	0.79	**0.75**	0.71	0.68	0.75	0.71	**0.68**	0.65	0.62	0.71	0.68	**0.65**	0.62	0.59	0.69	0.65	**0.62**	0.60	0.57	0.66	0.63	**0.60**	0.57	0.55
Tryptophan, %	0.25	0.24	**0.23**	0.22	0.21	0.23	0.22	**0.21**	0.20	0.19	0.22	0.21	**0.20**	0.19	0.18	0.21	0.20	**0.19**	0.18	0.18	0.20	0.19	**0.18**	0.18	0.17
Arginine, %	1.08	1.02	**0.97**	0.93	0.88	0.97	0.93	**0.88**	0.85	0.81	0.93	0.88	**0.84**	0.81	0.77	0.90	0.85	**0.81**	0.78	0.75	0.86	0.82	**0.78**	0.75	0.72
Isoleucine, %	0.85	0.81	**0.77**	0.73	0.70	0.76	0.72	**0.69**	0.66	0.63	0.71	0.68	**0.65**	0.62	0.60	0.69	0.66	**0.63**	0.60	0.57	0.66	0.63	**0.60**	0.58	0.55
Valine, %	0.96	0.91	**0.86**	0.82	0.79	0.85	0.81	**0.78**	0.74	0.71	0.81	0.77	**0.73**	0.70	0.67	0.78	0.74	**0.71**	0.67	0.65	0.75	0.71	**0.68**	0.65	0.62
Crude protein[6], %	20.78	19.68	**18.70**	17.81	17.00	18.70	17.81	**17.00**	16.26	15.58	17.80	16.95	**16.18**	15.48	14.83	17.20	16.38	**15.64**	14.96	14.33	16.50	15.71	**15.00**	14.35	13.75
Sodium, %	0.20	0.19	**0.18**	0.17	0.16	0.17	0.16	**0.15**	0.15	0.14	0.17	0.16	**0.15**	0.15	0.14	0.17	0.16	**0.15**	0.15	0.14	0.17	0.16	**0.15**	0.15	0.14
Chloride, %	0.20	0.19	**0.18**	0.17	0.16	0.17	0.16	**0.15**	0.15	0.14	0.17	0.16	**0.15**	0.15	0.14	0.17	0.16	**0.15**	0.15	0.14	0.17	0.16	**0.15**	0.15	0.14
Linoleic acid (C18:2 n-6), %	2.22	2.11	**2.00**	1.90	1.82	2.00	1.90	**1.82**	1.74	1.67	1.60	1.52	**1.45**	1.39	1.33	1.50	1.43	**1.36**	1.30	1.25	1.40	1.33	**1.27**	1.22	1.17
Choline, mg/kg	1778	1684	**1600**	1524	1455	1800	1714	**1636**	1565	1500	1800	1714	**1636**	1565	1500	1800	1714	**1636**	1565	1500	1800	1714	**1636**	1565	1500

CALCIUM AND PHOSPHORUS CHANGES BASED ON FEED INTAKE																									
	Weeks 18–33							Weeks 34–48					Weeks 49–62					Weeks 63–76					Weeks 77+				
Feed Consumption, g/day per bird	90	95	100	**105**	110	115	120	100	105	**110**	115	120	100	105	**110**	115	120	100	105	**110**	115	120	100	105	**110**	115	120
Calcium[7,8], %	4.44	4.21	4.00	**3.81**	3.64	3.48	3.33	4.20	4.00	**3.82**	3.65	3.50	4.40	4.19	**4.00**	3.83	3.67	4.60	4.38	**4.18**	4.00	3.83	4.70	4.48	**4.27**	4.09	3.92
Phosphorus (available)[7,9], %	0.48	0.46	0.43	**0.41**	0.39	0.38	0.36	0.41	0.39	**0.37**	0.35	0.34	0.37	0.36	**0.34**	0.32	0.31	0.35	0.33	**0.32**	0.30	0.29	0.32	0.31	**0.29**	0.28	0.27
Phosphorus (digestible), %	0.43	0.41	0.39	**0.37**	0.35	0.34	0.32	0.37	0.35	**0.33**	0.32	0.31	0.34	0.32	**0.31**	0.29	0.28	0.31	0.30	**0.29**	0.27	0.26	0.29	0.28	**0.26**	0.25	0.24

[1] All nutrient requirements are based on the Feed Ingredient Tables.

[2] Crude protein, methionine+cystine, fat, linoleic acid, and / or energy may be changed to optimize egg size.

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[4] A good approximation of the influence of temperature on energy needs is that for each 0.5°C change higher or lower than 22°C, subtract or add about 2 kcal /bird /day, respectively.

[5] Recommendation for Total Amino Acids is only appropriate to corn and soybean meal diet. Where diets utilize other ingredients, recommendations for Standardized Ileal Digestible Amino Acids must be followed.

[6] Diets should always be formulated to provide required intake of amino acid. Concentration of crude protein in diet will vary with raw material used. Crude protein value provided is an estimated typical value only.

[7] Calcium and available phosphorus requirements are determined by flock age. When production remains higher and diets are fed for longer than ages shown, it is recommended to increase to calcium and phosphorus concentrations of next feeding phase.

[8] Calcium carbonate particle size recommendation varies throughout lay. Refer to Calcium Particle Size. Dietary calcium levels and fine to coarse ratio may need to be adjusted based on limestone solubility.

[9] Where other phosphorus systems are used, diets should contain recommended minimum level of available phosphorus.

Vitamins and Trace Minerals

ITEM [1,2,3,4]	IN 1000 KG COMPLETE DIET	
	Rearing Period	Production Period
Vitamin A, IU	10,000,000	8,000,000
Vitamin D_3[5], IU	3,300,000	3,300,000
Vitamin E, g	30.00	25.00
Vitamin K (menadione), g	3.50	3.00
Thiamin (B_1), g	2.20	2.50
Riboflavin (B_2), g	6.60	5.50
Niacin (B_3)[6], g	40.00	30.00
Pantothenic acid (B_5), g	10.00	10.00
Pyridoxine (B_6), g	4.50	5.00
Biotin (B_7), mg	100.00	75.00
Folic acid (B_9), g	1.00	0.90
Cobalamine (B_{12}), mg	23.00	23.00
Manganese[7], g	100.00	100.00
Zinc[7], g	85.00	80.00
Iron[7], g	30.00	40.00
Copper[7], g	15.00	8.00
Iodine, g	1.50	1.20
Selenium[7], g	0.25	0.25

[1] Minimum recommendations for rearing and laying periods. Local regulations may limit dietary content of individual vitamins or minerals. Levels of 150-200mg/kg of Vitamin C can be beneficial during periods of stress.

[2] Store premixes according to supplier's recommendations and observe 'use by' dates to ensure vitamin activity is maintained. Inclusion of antioxidant may improve premix stability.

[3] Vitamin and mineral recommendations vary according to activity.

[4] Where heat treatment is applied to diet, higher levels of vitamins may be required. Consult with vitamin supplier regarding stability through individual production processes.

[5] A proportion of Vitamin D_3 can be supplemented as 25-hydroxy D_3 according to supplier's recommendations and applicable limits.

[6] Higher levels of Niacin are recommended in non-cage systems.

[7] Greater bioavailability and productivity may be possible with use of chelated mineral sources.

[8] Supplementing with up to 500 ppm of magnesium may be beneficial to support eggshell quality, particularly in aged hens or during periods of increased metabolic demand.

Drinking Water Quality for Poultry

ITEM	MAXIMUM CONCENTRATION (ppm or mg/L)*	
Nitrate NO_3^- [1]	25	Older birds will tolerate higher levels up to 20 ppm. Stressed or disease challenged birds may be more sensitive to effects of Nitrate.
Nitrate Nitrogen (NO_3-N) [1]	6	
Nitrite NO_2^- [1]	4	Nitrite is considerably more toxic than Nitrate, especially for young birds, where 1 ppm Nitrite may be considered toxic.
Nitrite Nitrogen (NO_2-N) [1]	1	
Total dissolved solids [2]	1000	Levels up to 3000 ppm may not affect performance but could increase manure moisture.
Chloride (Cl^-) [1]	250	Levels as low as 14 mg may be problematic if sodium is higher than 50 ppm.
Sulphate (SO_4^-) [1]	250	Higher levels may be laxative.
Iron (Fe) [1]	<0.3	Higher levels result in bad odour and taste.
Magnesium (Mg) [1]	125	Higher levels may be laxative. Levels above 50 ppm may be problematic if sulphate levels are high.
Potassium (K) [2]	20	Higher levels may be acceptable depending on sodium level, alkalinity, and pH.
Sodium (Na) [1,2]	50	Higher concentration is acceptable but concentrations above 50 ppm should be avoided if high levels of chloride, sulphate, or potassium exist.
Manganese (Mn) [3]	0.05	Higher levels may be laxative.
Arsenic (As) [2]	0.5	
Fluoride (F^-) [2]	2	
Aluminium (Al) [2]	5	
Boron (B) [2]	5	
Cadmium (Cd) [2]	0.02	
Cobalt (Co) [2]	1	
Copper (Cu) [1]	0.6	Higher levels result in bitter taste.
Lead (Pb) [1]	0.02	Higher levels are toxic.
Mercury (Hg) [2]	0.003	Higher levels are toxic.
Zinc (Zn) [1]	1.5	Higher levels are toxic.
pH [1]	5–7	Birds may adapt to lower pH. Below pH 5 may reduce water intake and corrode metal fittings. Above pH 8 may reduce intake and reduce effectiveness of water sanitation.
Total bacteria counts [3]	1000 CFU/ml	This is likely to indicate dirty water.
Total Coliform bacteria [3]	50 CFU/ml	
Faecal Coliform bacteria [3]	0 CFU/ml	
Oxygen Reduction Potential (ORP) [3]	650–750 mEq	The ORP range at which 2–4 ppm of free chlorine will effectively sanitise water at a favourable pH range of 5–7.

* Limits may be lower as interactions exist between magnesium and sulphate; and between sodium, potassium, chloride, and sulphate.

[1] Carter & Sneed, 1996. Drinking Water Quality for Poultry, Poultry Science and Technology Guide, North Carolina State University Poultry Extension Service. Guide no. 42

[2] Marx and Jaikaran, 2007. Water Analysis Interpretation. Agri-Facts, Alberta Ag-Info Centre. Refer to http://www.agric.gov.ab.ca/app84/rwqit for online Water Analysis Tool

[3] Watkins, 2008. Water: Identifying and Correcting Challenges. Avian Advice 10(3): 10–15 University of Arkansas Cooperative Extension Service, Fayetteville

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